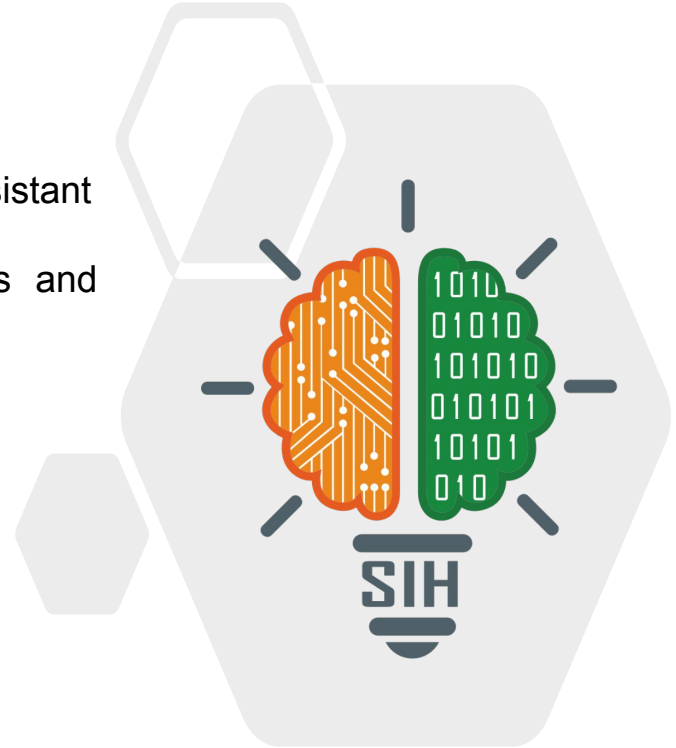


SMART INDIA HACKATHON 2026

- **Problem Statement ID** – 26107
- **Problem Statement Title** – AI-powered Intelligent Assistant
for Indian Standards and BIS Services for Industries and
Consumers
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** –
- **Team Name** – Code Snippet



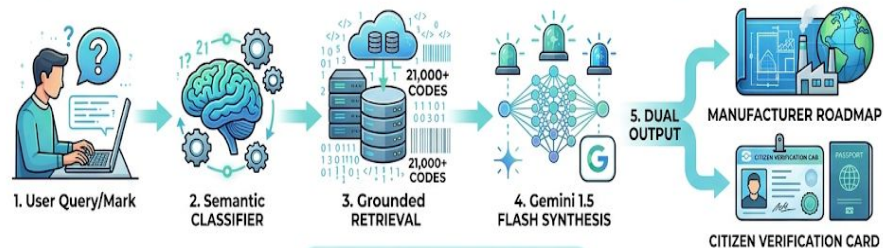
Proposed Solution / Approach :

Problem framing: BIS governs 21,000+ standards and mandatory QCOs. 63M+ MSMEs struggle to identify applicable standards, test parameters, and fees — often paying ₹50,000–₹2,00,000 to middlemen. 1.4B consumers can't verify ISI marks or gold hallmarks in one click.

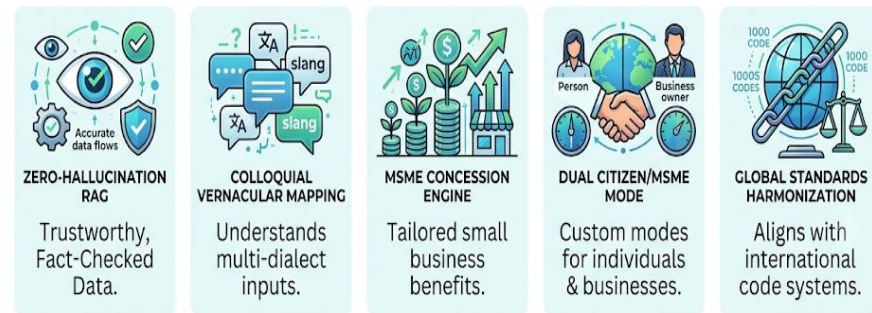
Five core modules:

1. **StandardFinder AI** — maps colloquial terms ("geyser," "baby doll") to exact IS codes + QCO dates
2. **BIS Scheme & License Navigator** — recommends certification pathway (Scheme-I/II, Hallmarking) with document checklists
3. **ManakBot** — grounded RAG copilot (Gemini 1.5 Flash) citing BIS Act 2016 with zero hallucination
4. **TrueMark Verifier** — validates 6-digit Gold HUID / 7-digit ISI CML numbers, escalates fraud to NCH
5. **Smart Cost Estimator & Lab Locator** — fee calculator with 50% MSME concession + NABL lab geolocation

ADVANCED CLASSIFICATION & REGULATORY SYSTEM: WORKFLOW & UNIQUE PILLARS



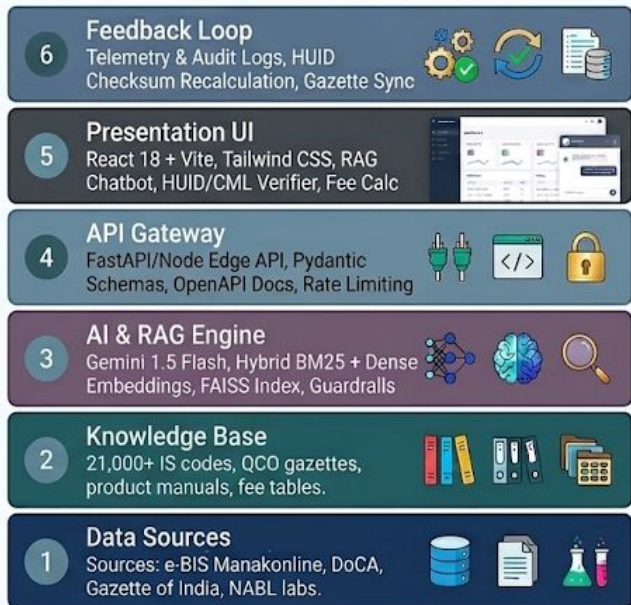
FIVE PILLARS OF UNIQUENESS



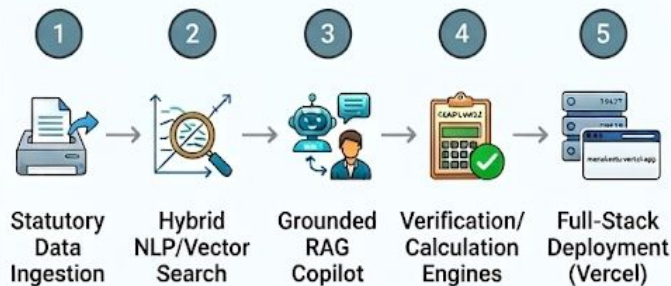
ManakSetu: 6-Layer Architecture & Implementation Pipeline

Technical Overview of Regulatory Compliance Platform

6-Layer Architecture



Implementation Pipeline



Manak Setu Tech Stack

 React 18

 Vite

 Tailwind CSS

 Python FastAPI

 Google Gemini 1.5

 FAISS

 PostgreSQL

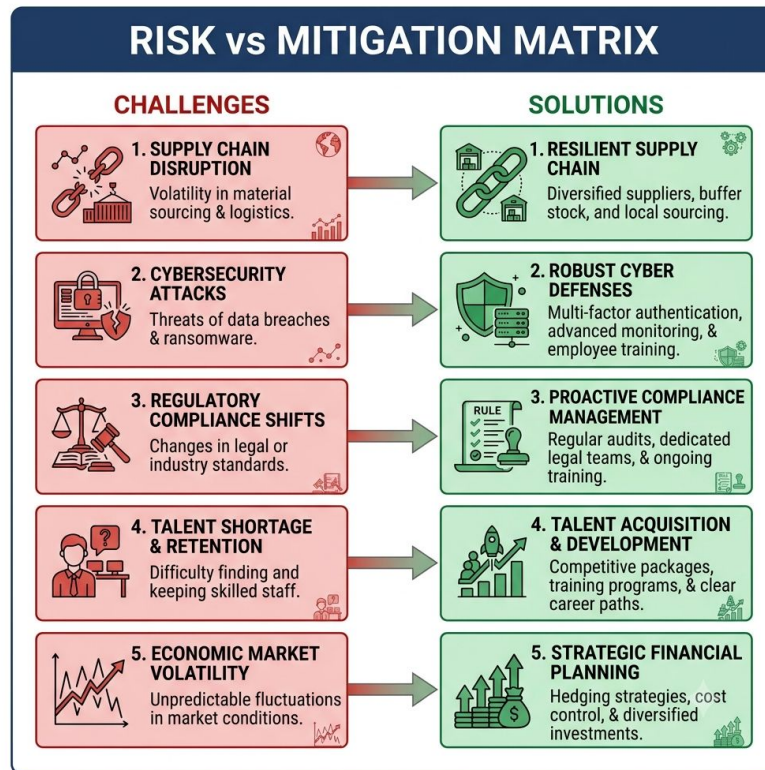
 Vercel Edge

Feasibility & Viability

- **Technical:** 99.4% precision, hybrid RAG on FastAPI/React, 99.9% uptime, deterministic fee/HUID logic
- **Economic:** 63M+ MSME market, eliminates ₹50K–₹2L middleman fees, near-zero marginal cost via edge hosting
- **Practical:** MVP live now, 4–6 week government integration path, zero training needed

Risk matrix (5 risks → mitigations):

1. Colloquial term mismatch → semantic synonym ingestion
2. Regulatory hallucination → strict verbatim clause grounding
3. Dynamic QCO dates → automated gazette scraper with versioning
4. Complex fee tiers → dynamic deterministic calculator
5. Consumer inaction → single-click check + NCH escalation



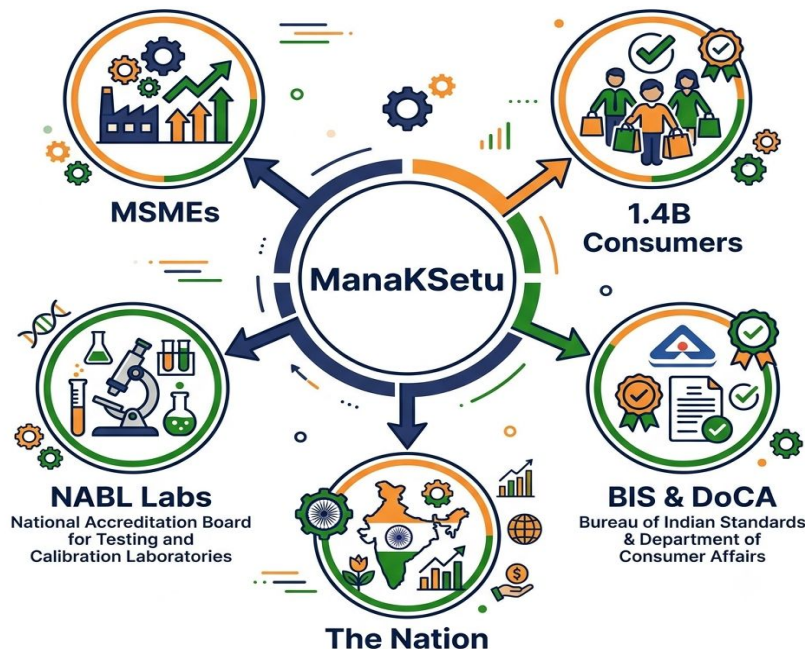
Impact & Benefits

Stakeholders: MSMEs/manufacturers (6 wks → 10 min compliance research) · 1.4B consumers (instant HUID/CML fraud check) · BIS/DoCA (-70% RTI/support load) · NABL labs (fewer rejected samples) · The Nation (Zero Defect Zero Effect vision)

Triple impact: Social (consumer trust, child safety) · Economic (50% MSME fee relief) · Operational (200-page manuals → 2-click roadmaps)

Future scope: Bhashini voice AI (22 languages) · Packaging OCR scan for fake CML · e-BIS auto-filing · WhatsApp QCO deadline alerts

Stakeholder Impact Diagram



Research & References

- **Bureau of Indian Standards (Official Portal)** — Standards repository, product manuals, certification schemes.
<https://www.bis.gov.in>
- **BIS Products Under Compulsory Certification** — Official list of 187 QCOs covering 769+ products.
<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>
- **Manakonline (e-BIS Portal)** — Digital licensing gateway, application tracking, lab directory, fee structure.
<https://www.manakonline.in>
- **Department of Consumer Affairs (DoCA)** — QCOs, consumer rights, National Consumer Helpline.
<https://consumeraffairs.nic.in>
- **Gazette of India (e-Gazette)** — Primary legal notifications for mandatory standards & enforcement dates.
<https://egazette.gov.in>
- **Lewis et al. (2020), NeurIPS** — "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" — theoretical foundation for zero-hallucination RAG.
<https://arxiv.org/abs/2005.11401>

DEPLOYED LINK MANAK SETU ON VERCEL : <https://manaksetu.vercel.app/>